

Project Proposal

AI Based Real-Time Speckle Imaging for Freshness Analysis

Project Guides:

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Project Summary:

This project presents an easy-to-use imaging system that can quickly assess the quality of fruits and other organic materials. With a simple setup, it provides real-time insights into freshness and subtle changes, making it a valuable tool for farmers, vendors, and consumers alike. By offering a cost-effective alternative to traditional methods, this system makes quality checks more accessible and helps ensure fresher produce, reduce food waste, and promote more sustainable agricultural practices.

Project Objectives:

- Implement a mobile application that can process and analyze speckle images in real time using AI and Machine learning techniques.
- Conduct experiments to validate the accuracy of the system in detecting freshness and physiological changes in agricultural produce.

- Provide a solution that can be used for agricultural quality assessment.

Items Summary for Portable Smartphone-Based Biospeckle System

1. He-Ne Laser: (Required)

A laser source for generating speckle patterns used in analyzing the surface of biological samples like fruits and vegetables.

Estimated Cost: Rs. 36,000

2. Beam Splitter: (Required)

Beam splitter can play an important role in improving the accuracy of a biospeckle imaging project.

Estimated Cost: Rs. 10,000

3. Concave lens: (Required)

A concave lens can be particularly useful in expanding the beam for uniform sample illumination, adjusting beam intensity, and controlling speckle size, which can contribute to better image capture and analysis of the biospeckle patterns.

Estimated Cost: Rs. 500 - 600

4. Tripod and Mounting Equipment: (Already existing in college)

Essential for stabilizing the smartphone to ensure clear and consistent image capture during the speckle analysis.

5. Experimental Sample Materials:

Includes fruits, vegetables, and other biological samples required for testing and validating the system's accuracy.

6. Miscellaneous Expenses:

Accounts for any additional equipment, power adapters, cables, and unforeseen costs that might arise during the setup.

7. Mobile Camera:

Used for capturing speckle images.

8. DSLR Camera: (Already existing in college)

High-quality camera for detailed image capture

9. Red Monochromatic Laser: (Already existing in college)

A laser source for generating speckle patterns used in analyzing the surface of biological samples like fruits and vegetables. **This laser is not effectively generating speckle images.**

Please note that this project is currently in the trial stage. While we are utilizing the He-Ne laser for speckle image generation, the required output and accuracy of the analysis may not be assured at this point. Ongoing testing and adjustments will be necessary to optimize performance and achieve reliable outcomes.

As an alternative, we are planning to seek permission to use the required equipment at Indian Institute of Science (IISc) Bangalore. This could provide us with access to more advanced resources, potentially enhancing the accuracy and reliability of our results.